Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If g is the inverse function of f and $f'(x) = \sin x$, then $g'(x)$ is

(A) $\operatorname{cosec}\{g(x)\}$

(B) $\sin\{g(x)\}$

(C) $-\frac{1}{\sin\{g(x)\}}$

(D) $\cos\{g(x)\}$

Q.2

If $f(x) = \sqrt[3]{\frac{x^4}{|x|}}, x \neq 0$ and $f(0) = 0$ is:

(A) continuous for all x but not differentiable for any x

(B) continuous and differentiable for all x

(C) continuous for all x and differentiable for all $\mathrm{x} \neq 0$

(D) continuous and differentiable for all $x \neq 0$

Q.3

If the equation $a_n x^n + a_{n-1} x^{n-1} + \ldots\ldots\ldots\ldots + a_1 x = 0$ $\mathrm{a}_1 \neq 0, \mathrm{n} \geq 2$, has a positive root $\mathrm{x} = \alpha$, then the equation $\mathrm{na_n x^{n-1}} + (\mathrm{n}-1)\mathrm{a_{n-1}x^{n-2}} + \ldots\ldots\ldots + \mathrm{a_1} = 0$ has a positive root, which is

(A) greater than α

(B) smaller than α

(C) greater than or equal to α

(D) equal to α

Q.4

If $\sqrt{1-x^{2n}} + \sqrt{1-y^{2n}} = a\left(x^n - y^n\right)$, then $\sqrt{\frac{1-x^{2n}}{1-y^{2n}}}\frac{dy}{dx}$ is equal to

(A) 1

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(B) x/y

(C) $\frac{x^{n-1}}{y^{n-1}}$

(D) None of these

Q.5

The equation $e^{x-8} + 2x - 17 = 0$ has

(A) two real roots

(B) one real root

(C) eight real roots

(D) four real roots

Q.6

Choose the correct statements $-$

(A) If $f'(a^+)$ and $f'(a^-)$ exist finitely at a point, then f is continuous at $x = a$.

(B) The function $f(x) = 3\tan 5x - 7$ is differentiable at all points in its domain.

(C) The existence of $\lim_{x\to c}(f(x) + g(x))$ does not imply of existence of $\lim_{x\to c} f(x)$ and $\lim_{x\to c} g(x)$

(D) All of these

Q.7

Let f be a function which is continuous and differentiable for all real x. If $f(2) = -4$ and $f'(x) \geq 6$ for all $x \in [2, 4]$, then

(A) $f(4) < 8$

(B) $f(4) \geq 8$

(C) $f(4) \geq 12$

(D) None of these

Q.8

If $f(0) = 0, f'(0) = 2$, then the derivative of $y = f(f(f(f(x))))$ at $x = 0$ is

(A) 2

(B) 8

(C) 16

(D) 4

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Q.9

If $y = \log_{10} x + \log_x 10 + \log_x x + \log_{10} 10$ then what is $\left(\frac{dy}{dx}\right)_{x=10}$ equal to?

(A) 10

(B) 2

(C) 1

(D) 0

Q.10

Which one of the following functions 1 s differentiable for all real values of x?

(A) $\frac{x}{|x|}$

(B) $x|x|$

(C) $\frac{1}{|x|}$

(D) $\frac{1}{x}$

Q.11

Let $0 < x < \pi$ and $y(x)$ be given by $(1 + \sin x)y^3 - (\cos x)y^2 + 2(1 + \sin x)y - 2\cos x = 0$ The derivative of y with respect to $\tan \frac{x}{2}$ at $x = \frac{\pi}{2}$ is :

(A) $\frac{1}{2}$

(B) $-\frac{1}{2}$

(C) 2

(D) -2

Q.12

If $y = \log^n x$, where $\log^n$ means $\log \log \log \ldots$ (repeated n time), then $x \log x \log x \log^2 x \log^3 x$ $\log^{n-1} x \log^n x \frac{dy}{dx}$ is equal to

(A) $\log x$

(B) $\log^n x$

(C) $\frac{1}{\log x}$

(D) 1

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Q.13

If $f(x) = \begin{cases} \frac{x \log \cos x}{\log(1+x^2)} & , x \neq 0 \\ 0 & , x = 0 \end{cases}$ then $f(x)$ is

(A) continuous as well as differentiable at $\mathrm{x} = 0$

(B) continuous but not differentiable at $\mathrm{x} = 0$

(C) differentiable but not continuous at $\mathrm{x} = 0$

(D) neither continuous nor differentiable at $\mathrm{x} = 0$

Q.14

The number of points at which the function $\mathrm{f(x)} = |\mathrm{x} - 0.5| + |\mathrm{x} - 1| + \tan \mathrm{x}$ does not have a derivative in the interval $(0, 2)$ is

(A) 0

(B) 1

(C) 2

(D) 3

Q.15

Given $f : [-2a, 2a] \to R$ is an odd function such that the left hand derivative at $\mathrm{x} = \mathrm{a}$ is zero and $f(x) = f(2a - x) \forall x \in (a, 2a)$, then its left had derivative at $x = -a$ is

(A) 0

(B) a

(C) $-a$

(D) does not exist

Q.16

If $fx = \begin{cases} \frac{\sqrt{x}}{\sqrt{4+\sqrt{x}} - a} & x > 0 \\ c & x = 0 \\ \frac{4e^{\frac{2}{x}} + 3e^{\frac{1}{x}}}{e^{\frac{2}{x}} + be^{\frac{1}{x}}} & x < 0 \end{cases}$ is continuous at $x = 0$ for some constants a, b and c, then the value of $\frac{50b}{a}$ is equal to

(A) 18.75

(B) 45.22

(C) 15.85

(D) 14.85

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Q.17

Let $f(x) = \frac{(e^x - 1)^2}{\sin\left(\frac{x}{a}\right)\log\left(1+\frac{x}{4}\right)}$ for $x \neq 0$, and $f(0) = 12$. If f is continuous at $x = 0$, then the value of a is equal to

(A) 1

(B) -1

(C) 2

(D) 3

Q.18

If $fx = \begin{matrix} \frac{2^x - 1^2 \tan 3x}{x\sin^2 x} \\ \lambda \end{matrix} \begin{matrix} : \\ : \end{matrix} \begin{matrix} 0 < x < \frac{\pi}{6} \\ x = 0 \end{matrix}$ is continuous at $x = 0$, then the value of $\frac{10\sqrt{3\lambda}}{\ln 2}$ is equal to

(A) 30

(B) 40

(C) 50

(D) 60

Q.19

Which one of the following is correct in respect of the function $f(x) = \frac{x^2}{|x|}$ for $x \neq 0$ and $f(0) = 0$?

(A) $f(x)$ is discontinuous every where

(B) $f(x)$ is continuous every where

(C) $f(x)$ is continuous at $x = 0$ only

(D) $f(x)$ is discontinuous at $x = 0$ only

Q.20

Which one of the following is correct in respect of the function $f(x) = |x| + x^2$

(A) $f(x)$ is not continuous at $x = 0$

(B) $f(x)$ is differentiable at $x = 0$

(C) $f(x)$ is continuous but not differentiable at $x = 0$

(D) None of the above

Answers & Solutions

Q.1 Answer:

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$\operatorname{cosec}\{g(x)\}$

Solution:

Given $f^{-1}(x) = g(x)$

$\Rightarrow x = f[g(x)]$

Diff. both side w.r.t (x)

$\Rightarrow 1 = f'[g(x)] \cdot g'(x) \Rightarrow g'(x) = \frac{1}{f'(g(x))}$

Given, $f'(x) = \sin x \quad \therefore f'(g(x)) = \sin[g(x)]$

$\Rightarrow \frac{1}{f'(g(x))} = \operatorname{cosec}[g(x)]$

Hence, $g'(x) = \operatorname{cosec}[g(x)]$

Q.2 Answer:

continuous for all x and differentiable for all $\mathrm{x} \neq 0$

Solution:

$f(x) = \sqrt[3]{\frac{x^4}{|x|}}, x \neq 0, f(0) = 0$

$\therefore f(x) = \sqrt[3]{\frac{x^4}{-x}} = \sqrt[3]{-x^3} = -x$ **if** $x < 0$

$\& f(x) = \sqrt[3]{\frac{x^4}{x}} = \sqrt[3]{x^3} = x$ **if** $x > 0$

$$f(x) = \begin{cases} -x, & \text{if } x < 0 \\ 0, & \text{if } x = 0 \\ x, & \text{if } x > 0 \end{cases}$$

Clearly $\mathrm{f(x)}$ **is continuous for all** x **but not differentiables at** $x = 0$

Q.3 Answer:

smaller than α

Solution:

Let $f(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots\ldots\ldots\ldots + a_1 x = 0$

The other given equation, $\mathrm{na_n x^{n-1} + (n-1)a_{n-1} x^{n-2} + \ldots + a_1 = 0 = f'(x)}$

Given $\mathrm{a_1 \neq 0 \Rightarrow f(0) = 0}$

Again $\mathrm{f(x)}$ **has root** α, $\Rightarrow \mathrm{f(\alpha) = 0}$

$\therefore \mathrm{f(0) = f(\alpha)}$

$\therefore$ **By Rolle's theorem,** $\mathrm{f'(x) = 0}$ **has root between** $(0, \alpha)$

Hence $\mathrm{f'(x)}$ **has a positive root smaller than** α**.**

Q.4 Answer:

$\frac{x^{n-1}}{y^{n-1}}$

Solution:

Put $x^n = \cos\alpha, y^n = \cos\beta$

$\Rightarrow a = \frac{\sin\alpha + \sin\beta}{\cos\alpha - \cos\beta} = \frac{2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)}{-2\sin\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)}$

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$= -\cot\left(\frac{\alpha-\beta}{2}\right)$

$\Rightarrow 2\cot^{-1}(-a) = \alpha - \beta$

$\Rightarrow \cos^{-1}(x^n) - \cos^{-1}(y^n) = 2\cot^{-1}(-a)$

$\Rightarrow \frac{y^{n-1}}{\sqrt{1-y^{2n}}}\frac{dy}{dx} = \frac{x^{n-1}}{\sqrt{1-x^{2n}}} \Rightarrow \sqrt{\frac{1-x^{2n}}{1-y^{2n}}\frac{dy}{dx}} = \frac{x^{n-1}}{y^{n-1}}$

Q.5 Answer:

one real root

Solution:

$\text{Clearly} x = 8$ **satisfies the given equation. Assume that** $f(x) = e^{x-8} + 2x - 17 = 0$ **has a real root** α **other than** $x = 8$**. We may suppose that** $\alpha > 8$ **(the case for** $\alpha < 8$ **is exactly similar). Applying Rolle's theorem on** $[8, \alpha]$**, we get** $\beta \in (8, \alpha)$**, such that** $\mathrm{f}'(\beta) = 0$
But $\mathrm{f}'(\beta) = \mathrm{e}^{\beta-8} + 2$**, so that** $\mathrm{e}^{\beta-8} = -2$ **which is not possible, Hence there is no real root other than 8 .**

Q.6 Answer:

All of these

Solution:

$\lim_{\mathrm{h}\to 0^+} \frac{\mathrm{f(a+h)-f(a)}}{\mathrm{h}}$ **exist finitely**

$\therefore \lim_{\mathrm{h}\to 0^+} \mathrm{f(a+h)-f(a)}$

$$\Rightarrow \lim_{\mathrm{h}\to 0^+} \left(\frac{\mathrm{f(a+h)-f(a)}}{\mathrm{h}}\right)\mathrm{h} = 0$$

$$\text{Similarly, } \lim_{\mathrm{h}\to 0^-} \mathrm{f(a+h)} = \mathrm{f(a)}$$

$\therefore \mathrm{f}$ **is continuous at** $\mathrm{x} = \mathrm{a}$

(b) Function is not differentiable at $5x = (2n+1)\frac{\pi}{2}$ **only, which are not in domain**

(c) Let $f(x) = \frac{1}{x^2}$ **and** $g(x) = -\frac{1}{x^2}$**,**

$\lim_{x\to 0} f(x) + g(x)$ **exists whatever** $\lim_{x\to 0} f(x)$

and

$\lim_{x\to 0} g(x)$ **does not exist.**

Q.7 Answer:

$f(4) \geq 8$

Solution:

By mean value theorem, there exists a real number $c \in (2,4)$ **such that** $f'(c) = \frac{f(4)-f(2)}{4-2} \Rightarrow f'(c) = \frac{f(4)+4}{2}$

Since, $f'(c) \geq 6, \forall x \in [2,4]$

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$\therefore f'(c) \geq 6 \Rightarrow \frac{f(4)+4}{2} \geq 6$
$\Rightarrow f(4)+4 \geq 12 \Rightarrow f(4) \geq 8$

Q.8 Answer:

16

Solution:

$$\begin{aligned} y'(x) &= f'(f(f(f(x))))f'(f(f(x)))f'(f(x))f'(x) \\ \Rightarrow y'(0) &= f'\left(f(f(f(0))\right)\left(f'(f(f(0)))f'\left(f'(0)\right)f'(0) \\ &= f'(f(f(0)))f'(f(0))f'(0)f'(0) \\ &= f'(f(0))f'(0)f'(0)f'(0) \\ &= f'(0)f'(0)f'(0)f'(0) = \left(f'(0)\right)^4 = 2^4 = 16. \end{aligned}$$

Q.9 Answer:

0

Solution:

$y = \log_{10} x + \log_x 10 + \log_x x + \log_{10} 10$
$y = \log_{10} x + \log_x 10 + 1 + 1$
Differentiating equation w.r.t. x $\frac{dy}{dx} = \frac{1}{x \log_e 10} - \frac{1}{(\log_{10} x)^2} \cdot \frac{1}{(x \log 10)}$
$= \frac{1}{x \log_e 10}\left[1 - \frac{1}{(\log_{10} x)^2}\right]$
$\left(\frac{dy}{dx}\right)_{x=10} = \frac{1}{10 \log_e 10}[1-1] = 0$

$$\left[\begin{array}{rl} \text{Note: } \log_x 10 & = \frac{\log_{10} 10}{\log_{10} x} = \frac{1}{\log_{10} x} \\ \frac{d}{dx}\left[\frac{1}{\log_{10} x}\right] & = -\left(\log_{10} x\right)^{-2} \times \frac{1}{x \log_e 10} \\ & = -\frac{1}{(\log_{10} x)^2 x \log_e 10} \end{array}\right]$$

Q.10 Answer:

$x|x|$

Solution:

Since $\frac{x}{|x|}$ **is not continuous function**
$\therefore$ it is not differentiable also. Also, L.H.D. and R.H.D. at $x = 0$ **not equal. Thus, only function given in option '** b **' gives differentiability for all real values of** x**.**

Q.11 Answer:

$-\frac{1}{2}$

Solution:

The given eq. can be written as $\left(y^2+2\right)\left(y - \frac{\cos x}{1+\sin x}\right) = 0$
$\Rightarrow y = \frac{\cos x}{1+\sin x} \quad \left[\because y^2+2 \neq 0\right]$

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$= \frac{\frac{1-\tan^2\frac{x}{2}}{1+\tan^2\frac{x}{2}}}{1+\frac{2\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}}} = \frac{1-t^2}{1+t^2+2t} = \frac{(1-t)(1+t)}{(1+t)^2}$

$= \frac{1-t}{1+t} = \frac{2}{1+t} - 1, \quad$ **where** $t = \tan\frac{x}{2}$

$\Rightarrow \mathrm{y}'(\mathrm{t}) = \frac{-2}{(1+\mathrm{t})^2}$. **At** $\mathrm{x} = \frac{\pi}{2}, \mathrm{t} = 1$

$\therefore \mathrm{y}'(1) = \frac{-2}{(1+1)^2} = -\frac{1}{2}$

Q.12 Answer:

$\log^n x$

Solution:

$\because y = \log^n x$

On differentiating w.r.t. x**, we get** $x \log x \log^2 x \log^3 x \ldots \log^{n-1} x \log^n x \frac{dy}{dx}$

$= \frac{x \log x \log^2 x \log^3 x \ldots \log^{n-1} x \log^n x.1}{x \log x \log^2 x \log^3 x \ldots \log^{n-1} x}$

$= \log^n x$

Q.13 Answer:

continuous as well as differentiable at $\mathrm{x} = 0$

Solution:

We have,

$\mathrm{Lf}'(0) = \lim_{\mathrm{h}\to 0} \frac{\mathrm{f}(0-\mathrm{h})-\mathrm{f}(0)}{-\mathrm{h}} = \lim_{\mathrm{h}\to 0} \frac{-\mathrm{h}\log\cosh}{-\mathrm{h}\log(1+\mathrm{h}^2)}$

$= \lim_{\mathrm{h}\to 0} \frac{\log\cosh}{\log(1+\mathrm{h}^2)} \quad \left(\frac{0}{0} \text{ form}\right)$

$= \lim_{\mathrm{h}\to 0} \frac{-\tan\mathrm{h}}{2\,\mathrm{h}/(1+\mathrm{h}^2)} = -1/2$

$\mathrm{Rf}'(0) = \lim_{\mathrm{h}\to 0} \frac{\mathrm{f}(0+\mathrm{h})-\mathrm{f}(0)}{\mathrm{h}} = \lim_{\mathrm{h}\to 0} \frac{\mathrm{h}\log\cosh}{\mathrm{h}\log(1+\mathrm{h}^2)}$

$= \lim_{\mathrm{h}\to 0} \frac{\log\cos\mathrm{h}}{\log(1+\mathrm{h}^2)} \quad \left(\frac{0}{0} \text{ form}\right)$

$= \lim_{\mathrm{h}\to 0} \frac{-\tan\mathrm{h}}{2\,\mathrm{h}/(1+\mathrm{h}^2)} = \frac{-1}{2}$

Since $\mathrm{Lf}'(0) = \mathrm{Rf}'(0)$**, therefore** $\mathrm{f}(\mathrm{x})$ **is differentiable at** $\mathrm{x} = 0$ **Since differentiability** $\Rightarrow$ **continutity, therefore** $\mathrm{f}(\mathrm{x})$ **iscontinuous at** $x = 0$.

Q.14 Answer:

3

Solution:

$\mathrm{x} - \mathrm{a}\,|$ **is not differentiable at** $\mathrm{x} = \mathrm{a}$**. Also** $\tan\mathrm{x}$ **is not differentiable if**

$\mathrm{x} = (2\mathrm{k}+1)\frac{\pi}{2}, \mathrm{k} \in \mathbf{I}$

$\therefore$ **In the interval** $(0, 2), \mathrm{f}(\mathrm{x})$ **is not derivable at**

$\mathrm{x} = 0.5, \mathrm{x} = 1$ **and** $\mathrm{x} = \frac{\pi}{2}$

Q.15 Answer:

0

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Solution:

Given $f'(a) = \lim_{h\to 0} \frac{f(a-h)-f(a)}{-h} = 0 \ldots (1)$

Now $f'(-a^-) = \lim_{h\to 0} \frac{f(-a-h)-f(-a)}{-h}$

$= \lim_{h\to 0} \frac{-f(a+h)+f(a)}{-h}$

[$\because f(x)$ **is odd function** $= \lim_{h\to 0} \frac{-f(a-h)+f(a)}{-h}$

$[\because f(2a-x) = f(x) \Rightarrow f(a+x) = f(a-x)]$

$= \lim_{h\to 0} \frac{f(a-h)-f(a)}{h} = 0 \quad [\text{From}(1)]$

Q.16 Answer:

18.75

Solution:

In $\lim_{x\to 0^+} \frac{\sqrt{x}}{\sqrt{4+\sqrt{x}} - a}$ **form** $\frac{0}{0}$ **for** $a = 2$

$\Rightarrow \lim_{x\to 0^+} \frac{\sqrt{x}}{\sqrt{4+\sqrt{x}} - 2} \times \frac{\sqrt{4+\sqrt{x}}+2}{\sqrt{4+\sqrt{x}}+2}$

$\Rightarrow \lim_{x\to 0^+} \frac{\sqrt{x}\sqrt{4+\sqrt{x}}+2}{4+\sqrt{x}-4} = \lim_{x\to 0^+} \sqrt{4+\sqrt{x}} + 2 = 4$

Now, $\lim_{x\to 0^-} fx = \lim_{x\to 0^-} \frac{4e^{\frac{2}{x}} + 3e^{\frac{1}{x}}}{e^{\frac{2}{x}} + be^{\frac{1}{x}}}$

$= \lim_{x\to 0^-} \frac{e^{\frac{1}{x}}4e^{\frac{1}{x}}+3}{e^{\frac{1}{x}}e^{\frac{1}{x}}+b} = \lim_{x\to 0^-} \frac{4e^{\frac{1}{x}}+3}{e^{\frac{1}{x}}+b} = \frac{0+3}{0+b}$

For fx **to be continuous at** $x = 0$

$4 = \frac{3}{b} \Rightarrow b = \frac{3}{4}$

Therefore, $\frac{50b}{a} = \frac{50\times\frac{3}{4}}{2} = \frac{75}{4} = 18.75$

Q.17 Answer:

3

Solution:

$\text{Lt}_{x\to 0} \frac{(e^x-1)^2}{\sin\left(\frac{x}{a}\right)\log\left(1+\frac{x}{4}\right)}$

$= \frac{x}{a} \cdot \frac{\frac{(e^x-1)^2}{x}\cdot x^2}{\left(\frac{x}{a}\right)} \cdot \frac{\log\left(1+\frac{x}{4}\right)}{\frac{x}{4}} \cdot \frac{x}{4} \Rightarrow 4a = 12$

$x\to 0$

$a = 3$

Q.18 Answer:

30

Solution:

$\because$ fx **is continuous at** $x = 0$

$\therefore \lim_{x\to 0} fx = f0$

$\because \lim_{x\to 0^+} \frac{2^x - 1^2\tan 3x}{x.\sin^2 x} = \lambda$

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$\because \lim_{x \to 0^+} \frac{2^x - 1}{x}^2 \frac{\tan 3x}{3x} \cdot \frac{3x}{x} \frac{x^2}{\sin^2 x} = \lambda$

$\ln 2^2 . 1.3 . 1 = \lambda$

$\frac{10\sqrt{3\lambda}}{\ln 2} = 30$

Q.19 Answer:

$f(x)$ is continuous every where

Solution:

$$f(x) = \begin{cases} \frac{x^2}{x}, & x \neq 0 \\ 0 & x = 0 \end{cases}$$

$$= \begin{cases} \frac{x^2}{x} = x, & x > 0 \\ 0, & x = 0 \\ \frac{x^2}{-x} = -x, & x < 0 \end{cases}$$

Now, $\lim_{x \to 0^-} f(x) = \lim_{x \to 0}(-x) = 0$

$\lim_{x \to 0^+} f(x) = \lim_{x \to 0}(x) = 0$

and $f(0) = 0$ **So,** $f(x)$ **is continuous at** $x = 0$ **Also,** $f(x)$ **is continuous for all other values of** x**.**

Hence, $f(x)$ **is continuous everywhere.**

Q.20 Answer:

$f(x)$ is continuous but not differentiable at $x = 0$

Solution:

$$\Rightarrow f(x) = \begin{cases} x^2 + x, & x \geq 0 \\ x^2 - x, & x < 0 \end{cases}$$

$\text{LHL} = \lim_{x \to 0^-} f(x)$

$= \lim_{h \to 0} f(0 - h) = \lim_{h \to 0}(0 - h)^2 - (0 - h)$

$= \lim_{h \to 0} h^2 + h = 0$

and $\text{RHL} = \lim_{x \to 0} f(x) = \lim_{h \to 0} f(0 + h)$

$= \lim_{h \to 0}(0 + h)^2 + (0 + h)$

$= \lim_{h \to 0} h^2 + h = 0$

$\Rightarrow \text{LHL} = \text{RHL} = f(0)$

$\Rightarrow f(x)$ **is continuous at** $x = 0$ **Now, LHD** $= \lim_{h \to 0} \frac{f(0-h) - f(0)}{-h}$

$= \lim_{h \to 0} \frac{h^2 + h}{-h} = -\lim_{h \to 0} h + 1 = -1$

and, $\text{RHD} = \lim_{h \to 0} \frac{f(0+h) - f(0)}{h}$

$= \lim_{h \to 0} \frac{h^2 + h}{h} = \lim_{h \to 0} h + 1 = 1$

Thus, $\text{LHD} \neq \text{RHD}$

$\Rightarrow$ $f(x)$ **is not differentiable at** $x = 0$

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